INCH-POUND

MIL-O-55310/4E
15 March 1994
SUPERSEDING
MIL-O-55310/4D
4 October 1984

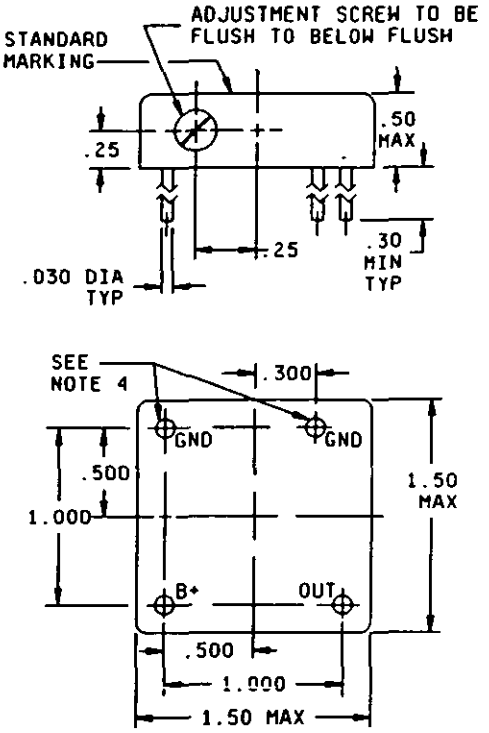
MILITARY SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 3 (TEMPERATURE COMPENSATED (TCXO)), 750 kHz THROUGH 10 MHz, SOLDER SEAL, SQUARE WAVE, CMOS

Inactive for new design

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310



Inches	mm
.030	0.76
.25	6.4
.300	7.62
.500	12.70
1.000	25.40
1.50	38.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Pin number and function to be displayed on side of oscillator.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Solder (nonhermetic).

Weight: 2 ounces maximum.

Construction technology: Discrete construction.

Components: In accordance with MIL-STD-454.

Calibration: Manufacturer and user nominal frequency calibrated.

Screening: In accordance with MIL-O-55310, class B.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Oscillator load: 10 kilohms ±5 percent shunted by a 15-picofarad ±5 percent capacitor.

Output waveform: Symmetrical square wave, CMOS compatible.

Supply voltage: See table I.

Input current: At designated supply voltage (see table I).

Output frequency: Frequency as designated at time of acquisition (see table I).

Initial accuracy at reference temperature: ±5 ppm maximum at +25°C ±3°C.

Frequency adjustment: (Mechanical, adjustable after the cover is on.) A tuning adjustment for aging shall have a range of ±3.5 ppm. Resolution shall be 1.0 Hz maximum. Grounding the tuning screw shall not damage the circuit.

Output voltage: Output voltage into a 10-kilohm ±5 percent load shunted by a 15-picofarad ±5 percent capacitor (see table I).

Rise and fall times: 50 ns maximum.

Duty cycle: 35 to 65 percent duty cycle.

Initial frequency-temperature accuracy (one-half temperature cycle): See table I. Measurements taken at 10 equispaced increments over the specified operating temperature range.

Frequency-voltage tolerance: ±1 ppm maximum for a ±0.25 volt change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

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TABLE I. Dash numbers and operating characteristics.

Dash number	Output frequency range	Supply voltage ±5%	Output voltage		Input current at designated supply voltage (max)	Operating temperature range 1/ 2/	Initial frequency-temperature accuracy
			Logic 1 (min)	Logic 0 (max)			
		V dc	Volts	Volts	mA		ppm
01	750 kHz to 6.3 MHz	+15	13	2	15	-55°C to +85°C	±5
02	6.3 MHz to 10 MHz	+15	13	2	20	-55°C to +85°C	±5
03	750 kHz to 6.3 MHz	+12	10	1	12	-55°C to +85°C	±5
04	6.3 MHz to 10 MHz	+12	10	1	15	-55°C to +85°C	±5
05	750 kHz to 6.3 MHz	+10	9.5	0.5	10	-55°C to +125°C	±10
06	6.3 MHz to 10 MHz	+10	9.5	0.5	12	-55°C to +125°C	±10

1/ -55°C to +85°C (designator letter D).
2/ -55°C to +125°C (designator letter A).

Frequency aging: Measurements shall be taken at +70°C ±0.2°C.

±0.1 ppm per 30 days, maximum.

±0.3 ppm per 90 days, maximum. 1/

±1 ppm per year, maximum. 1/

Terminal strength: MIL-STD-202, method 211, test condition A.

Applied force: 2 pounds each terminal.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal:

Nonoperating: In accordance with MIL-O-55310 and MIL-STD-202, method 204, test condition B.

Operating: In accordance with MIL-O-55310 and MIL-STD-202, method 204, test condition C.

Ambient pressure:

Nonoperating: In accordance with MIL-O-55310 except 20,000 feet maximum.

Operating: MIL-STD-202, method 105, test condition F.

Exposure time prior to measurement: 30 minutes.

1/ This is a performance requirement of the oscillator. Support data shall be presented showing that this requirement shall be met.

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QUALITY ASSURANCE PROVISIONS:

Group C inspection: Not applicable.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and code alphas, and numeric number. See example:

EXAMPLE

M55310/04- B 01 A XXXXXXXX

M prefix and specification sheet number

Product assurance level (B only)

Dash number (see table I)

Operating temperature range (A or D) (see table I)

Frequency

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0677-01)

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